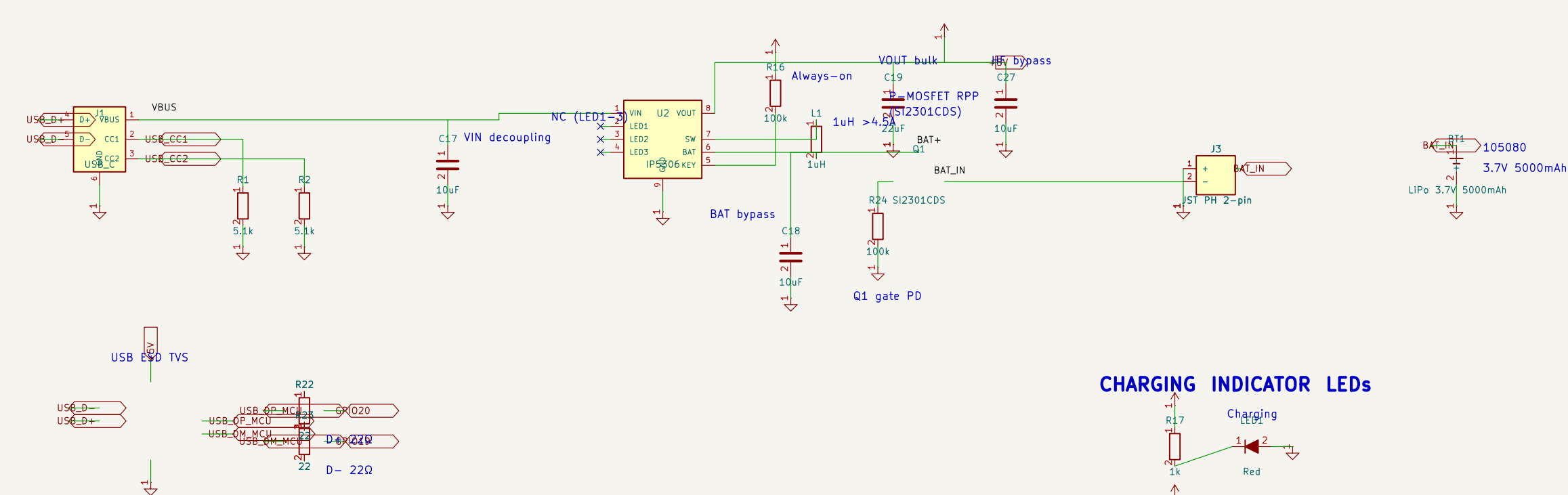
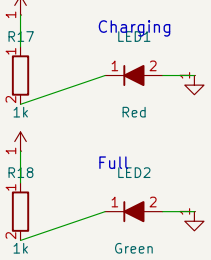


POWER SUPPLY

USB-C -> IP5306 SOP-8 (charge + boost) -> AMS1117-3.3 -> 3.3V rail

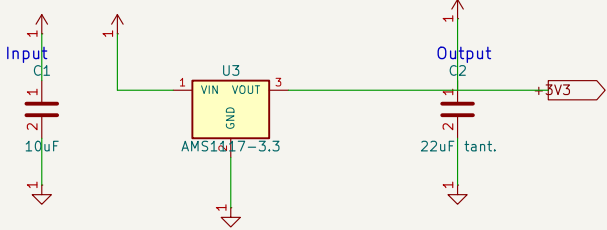


CHARGING INDICATOR LEDs

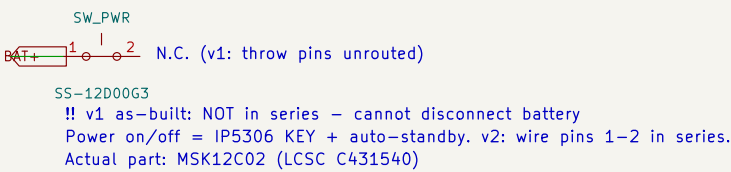


VOLTAGE REGULATOR

5V -> AMS1117-3.3 -> 3.3V (800mA max)



POWER SWITCH



Design Notes:

- IP5306 eSOP-8: integrated charger + synchronous boost (no ext. Schottky)
- 1uH inductor: BAT -> L1 -> SW (>4.5A saturation, shielded)
- GND via exposed pad only (must solder to ground plane)
- KEY pulled to +5V via 100k for always-on operation
- AMS1117-3.3: stable 3.3V for ESP32-S3 and peripherals (800mA)
- 5.1k CC pull-downs identify USB-C UFP (5V sink)
- Battery: LiPo 3.7V 5000mAh via JST PH connector